Debarghya Mukherjee

Roll No.: B124040 **Department:** Computer Science and Engineering
Degree: Bachelor of Technology **Institution:** International Institute of Information Technology, Bhubaneswar
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Education

- **International Institute of Information Technology, Bhubaneswar** *Aug 2024 – Jul 2028 (Expected)*
  Bachelor of Technology in Computer Science and Engineering CGPA: **8.00/10.0**
- **Aditya Birla Vani Bharati, West Bengal** Percentage: **85.3%**
  Senior Secondary (XII)
- **Gospel Home School, West Bengal** Percentage: **95.8%**
  Secondary (X)

Projects

- **AI Fitness Tracker** [GitHub]
  Full-stack, microservices-based fitness tracking platform with AI-driven activity recommendations.
  - Architected a **microservices system** with Spring Boot, using **Eureka** for service discovery, a centralized **Config Server**, and an **API Gateway** for routing.
  - Built independently deployable **User**, **Activity**, and **AI** services communicating asynchronously via **RabbitMQ**.
  - Integrated **Google Gemini AI** to generate personalized fitness and activity recommendations from user data.
  - Implemented secure **authentication and role-based authorization** using **Keycloak** (OAuth2 / OpenID Connect).
  - Designed a **polyglot persistence layer** — PostgreSQL for relational/user data and MongoDB for activity and AI recommendation logs.
  - **Containerized** all services with **Docker** for consistent, portable deployment.
  - Developed a **React.js** frontend for logging workouts and viewing AI-generated insights.
  - **Technologies Used:** Java, Spring Boot, Spring Cloud (Eureka, Gateway, Config Server), React.js, Docker, RabbitMQ, Keycloak, PostgreSQL, MongoDB, Gemini AI.
- **Royal Run** [GitHub]
  Fast-paced 3D arcade-style endless-runner game built in Unity.
  - Designed the core gameplay loop around coin collection, rare speed-boosting power-ups, and dynamic obstacle avoidance.
  - Implemented a **speed-management mechanic** requiring players to sustain velocity thresholds to reach checkpoints before time runs out.
  - Built a **progressive difficulty system** that scales obstacle density and flying-hazard frequency over time.
  - Programmed player movement, collision detection, and physics-based interactions using **Unity's C# scripting API**.
  - Structured the project with modular, reusable prefabs and scripts for obstacles, pickups, and checkpoints to enable rapid level iteration.
  - **Technologies Used:** Unity, C#, 3D Physics, Game Design.

Technical Skills and Interests

Languages: Java, C++, C#
CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS
Frameworks & Libraries: Spring Boot, Spring Cloud
Developer Tools: Docker, RabbitMQ, Keycloak, Git, GitHub
Databases: PostgreSQL, MongoDB
Areas of Interest: Backend Development, Microservices Architecture, Game Development

Achievements

- **Top 1.6%** in JEE Main among over 1 million candidates.
- Solved **380+ problems** on LeetCode, maintaining an active solving streak of **80+ days**.
- **Played and Captained** the **Straight Drive Cricket Academy** team (U13 and U15) in the **Cricket Association of Bengal (CAB)**.